

ActInf GuestStream 080.1 ~ Laura Desirée Di Paolo: "Active Inference Goes to School"

GuestStream | Laura Desirée Di Paolo | 1:24:35

all right

hello it is April 22nd 2024 we're in

active inference guest stream

80.1 with Laura desire Deo talking about

active inference Goes to School the

importance of active learning in the age

of llms we'll have a presentation than

some discussion so thank you Laura for

joining looking forward to

this thank you very much for having

invited me um okay so I'm going to

present practically one of our latest

publication which is um in fact that the

inference Goes to

School uh oh wait I have to Yes

um this is an overview of the talk more

or less I'll try to go as fast as I can

for the first part because I'd like to

spend a little bit more time on the

later slides and I'm sorry for some

misspelling or mistakes that you will

find here and there in the in the images

because I had long fights with Jud for

trying to uh um generate images that

would vate better my my my talk and at

some point I after 50 or 60 uh attempts

I simply gave gave up and they choose

just the best one that could I could uh

I could achieve to I could achieve okay

so ah still I have to use my

mouse uh just few things about myself

I've been uh interested in learning

since I uh uh I started University maybe

because I was uh considered a slow

learner in uh in high school definitely

not an high achiever uh um and therefore
when I started to uh to study at
University I started to study learning
different kind of learning different
learning across species and the
evolution of learning and so on and so
forth uh till the point in which I
started to be interested in learning in
educational settings thinking okay this
is the perfect recipe for Success
because you have all the ingredients
that you might have you have super
curious beings for whosoever had to deal
with children before they uh go to
school uh knows that they um have this
insane curiosity and they ask wise about
everything theoretically in school you
should have tons of learning
opportunities and you finally have
people that are trained to give uh the
answers that the children seek to
have unfortunately the sit the situation
in school is not always so nice for
quoting the um principal a British
principal and during an interview who
said it's incredible that children
prefer to be home sick that going to
school the place in which they can learn
stuff consider that learning is fun and
for what concerns me I am a researcher
and as I assume most of us um I love
learning I think learning very very fun
so I feel part of my my job or my duty
to bring back the fun into learning
particularly in educational settings
or in compulsory
education
um this is just a slide with some uh

basic literature about active inference
and learning I won't be too specific
about active inference uh there is here
a that eventually could um could follow
up for some questions more specific
about the active inference because it's
way more prepared than I am but anyway
this is like um

I don't know if you have I have to move
this one okay perfect um uh this is a
basic model of active uh inference the
idea of cognition in active inference
for which the agent and the cognitive
system of the agent is not waiting for
the information to arrive to fill in but
instead uh act proactivity so makes
predictions about the state of the world
in which expect to leave the information
that that expect to uh to perceive into
the world and then every time compare the
predictions that has made with the
effective information the effective
situation of the world in that
particular moment and when there is a
discrepancy discrepancy uh in in terms
of both surprise or events that are
unexpected or errors uh either update
the model for making better predictions
or change the words and so act onto the
world for uh the orinal
predictions in this model of cognition
learning is a sort of imperative because
every time that there is a discrepancy
between the prediction and the state of
the world is like a red flag that says
okay here there is something that can be
learned there is something new that can
um can be

acquired for making my future prediction
my future and error minimization
better um another interesting aspect of
deductive inference is that predictions
are not just in terms of thoughts
beliefs or knowledge for instance but
also in terms of sense of motor
activities so everything that we
perceive feel see um is considered a
prediction about the world and it's
considered in itself cognitive a way to
understand the world that surrounds
us the model of active inference um of
learning and cognition together is very
efficient and is uh very powerful for
explaining for instance different kinds
of uh learning or learning mechanism
using the same very simple so to say uh
mechanism you can um explain for
instance um individual trial and error
and imitation as every time the
generative model that
optimize on the environment and on the
situation the environment for acquiring
the the right information or for making
the the the best predictions possible um
you can also explain how expertise in
terms of refined knowledge what is
called know what uh or in terms of
skills uh can very easily come from
explorative policies and for Curious
Behavior into the world um
also importantly focuses on the role of
attention that if you think in terms of
educational settings and educational
environment is of course very important
again in terms of prediction error uh
Dynamics and surprise um for the

acquisition of competencies and skills and most importantly at least for me using the inference to uh process that very often are considered complementary but separated which are development and learning they can be Rec combined together because making predictions particularly in terms of sense of motor activity and sense of motor interaction with the word means also acting on our biology and neurobiological structure for being sure that these predictions are as accurate as they can be okay when we wrote said this when we wrote the paper we had three main problems the first problem was how do we apply a theory of cognition such as the active inference in real educational practices and conversely how these educational practices should look like for fitting our theory of cognition on these two problem adds up a third problem which is the uh the use and the spread of artificial intelligences particular L large language models in education as well as in other uh in other fields and how this might change the rules the problem of application of um large language models particular artificial intelligence or even other form of Technologies within educational settings but particularly uh large language models um is given uh mainly by the fact that they are disembodied when a user has to deal with a um larger language models in reality deals with

some artifacts the computer the the screen the the keyboard and so on that can touch see and perceive but at the end of the day um um Deals Only with the input output

series and the whole process all the other artifacts that are imprinted that are inside the uh the model itself are completely uh invisible um which makes a problem because normally we are used for instance we can't make Behavior reading or understanding intentionality behind the answers of uh of

chpt now I added uh yesterday morning this uh this slide this is a beautiful paper that came out I think two or three days ago about uh children the P the perception of children about information on Google and uh it's very interesting because it somehow it it has the same problem that

um that llms have uh children even very young children trust the information that Google uh gives but they don't have any um any possibility to understand where this C trust come from in the moment in which they have to deal with something like a chpt in which the the answer is given in a sort of natural language of course the problem is even uh is even

deeper uh therefore how do we nurture the human generative model in in the a in the in the age of the artificial

intelligences um these pictures that I have here on my slide I was looking around for uh different

representations of how integrate
artificial intelligences within
educational
settings uh and they all are accumulate
um they have one thing in common which
is they tend to treat the artificial
intelligence in in a sort of human way
because this is if the artificial
intelligences are disembodied um it
states that we are embodied embedded
extended and um enacted agent so we are
uh um um active inference agent so our
generative model for understanding for
dealing with these embodied entity
entities as somehow to um to put them in
a way that we can understand so our
problem would be how do we build an
educational environment in
which our generative models can
effectively and uh um efficaciously deal
with the powerfulness of these uh new
technologies one way is what I call here
the traditional way this is the only
image that I got from jpt that was
perfect at the first attempt I asked jpt
to give me the image of a classroom with
poop pills between eight and 10 years of
age um the fact that jpt gave me
directly this image tells me a lot
because jpt is trained on our human
knowledge and this is very likely the
image that we all have when we think
about a classroom about education this
that I called here traditional uh
traditional educational setting in
reality as a specific name which is the
teacher centered approach which I have
somehow schematize here um and more or

less works this way you have all the human knowledge that is far back in the background of the educational setting itself this extensive human knowledge is paralized and organized in curriculum that then affect the the child on the other hand you have a teacher that also affects the child and you have this child that somehow acquires these information and grows up having been filled by the information that is provided with um somehow scaringly this kind of uh structure is very similar to the kind of training that we do for disembodied entities such as llms with the additional uh problem that the fine tuning in educational setting in these kind of structures uh happens at different level not only the child is fine tuned by the teacher and by the curriculum having somehow knowledge that exist other knowledge outside of the classroom outside the knowledge that is provided within the educational settings uh but the fine-tuning works on the teachers themselves and the teachers themselves they don't fine tune every single child but they usually fine tune our old classroom now for whosoever knows how the fine tuning for instance in L works um knows that the more you f tune a model the least cre creative the model is going to be now um when we think about this kind of traditional settings uh and we think

that the active inference is the best um
uh model of cognition that we currently
have
uh we can't uh avoid to uh to uh
recognize that it works also in these
settings problem is yes it works but is
not really on point in the sense that um
for instance the uh um very often in
this kind of settings we pretend that
the attention of the poops goes
specifically to certain uh uh certain um
feature of the environment the the the
the lessons that are provided by the
teachers uh while in fact we know also
that children at particularly in early
age so till round about 12 they tend to
uh pay attention way more and to learn a
lot from irrelevant contextual
information which in active inference
terms makes a lot of sense because is a
generative model that that at uh till
that point in time very few experiences
so is gathering information in the
environment for making a better
prediction on the long
term another problem with this uh uh
with this model and this is the only
concept that is not in the paper that I
have added in the slide is what I called
here error learning Dynamics um another
problem with the this kind of settings
is what we consider the error signal we
have said before I have said before uh
in the moment in which there is a
discrepancy between the prediction that
the cognitive system has made and the
effect the actual uh um environmental uh
situation there is an error

signal if the error signal doesn't come exactly from the mistake but comes from for instance from the feedback that might arrive in terms of notes or from being scolded at home because you got a bad grade or you got a A not in school then of course the change of the state the change of the situation the acting onto the world and the update of the generative model and of the prediction H doesn't fit with the uh um uh with the uh with what we want the children to

learn uh these studies that I have here uh quoted some of them are really beautiful studies of Solange dero uh she has observed different children in different educational settings and also children and adults how do they they react to mistakes and how they monitor mistakes um and discovered that children educated in this kind of traditional or teacher centered uh approach uh educational settings develop more than uh the ability to learn from mistakes develop what is called error avoidance which again makes sense it's not that the the the active inference model doesn't work in this way it's only that the error signal doesn't come from the mistakes for instance in the task or in the exercise that the child has made but comes from another aspects of their world okay therefore how should the classrooms in which we want effectively to

Foster an idea of an active inference agent to nurture an active inference

agent um it has to be an educational setting it has to be a classroom that offer um um explorative opportunities and offer the time for precision waiting uh what I mean with that um it has to be a educational environment in which the um there is the possibility to learn from uh various aspects of of the world and there is there has to be the possibility for the for the child for the poil to um fine-tune its gener ative model uh multiple times on the same task in a way that can really really really uh understand um at best how to perform it it has to be an environment in which there are u in which it is fostered also a multimodal and multisensory way of approaching learning and and knowledge because as we said the um predictions are not just in terms of know what the the um um information that is provided to the child but also all the irrelevant stimuli that are around so irrelevant so they shouldn't be irrelevant should be instead relevant and has to Foster also individual interests and attention modalities because at the end of the day the generative model is always individual it's always the the individual that uh update its own generative model in its own specific environmental settings Al there are of course some coach forces as AEL would call them that makes us being able to um act and to think in in terms of our community of our

culture it has to be an educational environment in which development and learning are considered together and not separated in these traditional educational settings for for instance we have all the classroom that very often are divided by age not about skills or stage of development um it has to be an environment in which uh the construction of uh obstruction and categorization has to come through environmental experiences which again in active inference terms makes a lot of sense constructing abstract uh knowledge constructing abstract categories uh allows the system to be somehow prepared to how to deal with certain subject or with certain object or with certain circumstances uh and then requiring just the fine tuning of the of the generative model on the specificity of a particular circumstance and it has to uh to uh be an environment in which the ER learning Dynamics are considered in a way that the error signal comes directly from the error and not from from other um contextual uh information um instead of thinking to invent from the from the top of our mind a uh um a new educational approach we looked around to um search for which educational approach that's already out there that offered the offered the most opportunity to fit with our idea of educational settings uh among various the monor meod is the one that if

doesn't fit all the boxes uh
definitely fits most most of them um for
describing uh a little the in the Montessori
method you will always find a sort of open
spaces and multi-age uh environment of
learning curricula are always spanned on
free to six years um and not on just one
year as in um in traditional schools uh
children are always free to choose what
to study for how long uh with
whom uh it is already implemented
within the PED the pedagogy and within
the uh the theory also of Montessori
the idea that Learning and Development
are just one

process it works through what is called
an indirect form of teaching and through
the prepared environment prepared
environment is the classroom as is
called in the Montessori method um
which is already organized in a way that
can uh Foster some kind of not not not
only of skills but also of uh
construction of specific
knowledge and pays a lot of attention to
the error

Dynamics uh uh through the design of the
classroom itself but also through the
design of the specific learning material
I have some examples in in a couple of
slides um the materials in a way that
the material is designed so the
material is designed in a way that are
poor particularly in the
lower uh stages so in the in the first
grades uh can immediately perceive when
uh they have made a mistake and um Can
immediately correct them so the error

signals uh arrives directly it doesn't
there is no delay um and the old
classroom is organized in a way that the
uncertainty so the the um prediction
minimization is somehow manageable and
still organized in Progressive diff
difficulties but every time adding up
just one extra variable one extra
element of

uncertainty now is well known I wanton
if anyone wants to know more about the
the history of maror uh I can talk about
it later but it's well known that the
method was invented by maror Italian
physician at the uh in at the beginning
of last century

I have been uh lately dealing with the
most applied of our work particularly in
mathematics in geometry and in um
grammar and I have to admit that she was
frankly genial uh the the the design of
her material is is is astonishing still
today

um Deon is incredibly famous and
Incredibly well known there are more
than 15,000 montori School in the world
uh in more than 140 countries uh in
2011 um came out a paper uh in business
psychology I think it was called the
monor mafia because some of the richest
and most well-known entrepreneur in the
world um are orw montor

poils um but still it's not so much well
known the theories of development and
learning of the monor meod are not so
well known for instance in cognitive
Sciences so they are very much I was
very surprised the other day looking at

the how many results one can have
putting mono and entrepreneur uh in on
Google Scholars uh and not so much in
cognitive

Sciences this is not really related to
the topic but is also a good aspect of
the montor method which is given the
particular attention that gives to the
ecological context for learning it is
also it has also a leading position in
terms of ecological sustainability
within the educational setting and
within the curricula and this is of
course a topic of major interest that
should in my opinion at least be part of
whatever C in whatever School
independently of anything
else okay um these are some examples of
montesorri schools the montori schools
are more or less all structured in
similar

ways sorry uh in similar ways
um first and foremost there is no
distinction between inside and outside
the
classroom uh every space given the
physical constraints of the buildings uh
might be corridors might be um small or
big rooms the doors are always open the
children and the poils are always free
to move around there has to be always a
free a big free space available for uh
children that prefer to work on the
floor and also because some of the
materials requires lots of space there
is the material for history the history
band that's up to 5 m long and is
designed exactly for the children to

understand to have a a um a perceptual understanding of the different um positioning on the on in time of the different characters or events that they are working

on the inside and outside as I said they are both um educational into sense not only because there are lots of

activities like the observational nature and part of the curriculum of uh

botanics biology and so on that are literally run outside but because the

old Furnitures and uh the all everyday items are easily portable outside and very often um you might find class the classrooms or the normal uh daily work happening in in the outside

space uh there is one of the image that

I have here in in black and white uh

taken from from the B of from the book of Schneider and Mueller refers to a

mono School uh in Germany in the

30s um so it's historically designed in

this way uh within the montori classroom

also the construction of abstract uh

category ories and of categories in

general is um is supplied by the uh by

the environment itself uh through the

positioning of the objects of the

materials the books that refer to

specific uh disciplines of specific

subjects uh that are positioned in a way

to construct more and more abstract

categories for instance you might find

one might find the materials for uh

botany and materials for uh chemistry

and the materials for uh physics all

under the label

science and each one of them will be labeled as well in a way that children simply interacting in in with the space interacting with the material and with the material and simply with the simple activity of taking the material out from the shelf and put it in back understand more or less how the different subjects are related to one another Plus for instance through the presence of kitchens that are used for making food I used for uh experiments in chemistry I used for taking care of plants and I used for uh washing the brushes allows kids to uh make um connections across different disciplines um when I talked before about error learning Dynamics and how the the error signal should be um should be uh um should immediately get back to the system for having an effect an an effective uh update of the model or modification of the word uh in terms of uh effective learning um this is one example this is actually the first material for mathematics that is in the Mont meod that's usually presented to Children um between the age of 34 uh in Italian they are called the numerical roads in English they are called the red and blue uh roads and are these uh long pieces of of wood divided in a section that um uh represent units uh divid in different colors so in alternative colors blue and red um and

uh they are from 1 till till 10 with the
10 that's 1 meter long and the one
that's 10 cm long um these roads um have
the AIM to um help the the children to
um learn the basics of uh addition
additions and
subtractions um making perceptual
evidence evident uh how the different
quantities are related to one another
most importantly they don't have the AIM
to uh
um

teach the quantities because children of
three or four age they already ask for
free Bone bonds instead of two so they
already have an idea of quantities these
kind of material is designed exactly for
um uh uh making clear how quantities are
related to symbols and how quantities
are related to one

another this is the first of a set of
free materials for the uh for the the
beginning of the mathematical the
curriculum of mathematics which as a um
as a matter of fact has this
representational a the other two instead
that are the spindle B and the numerical
coin uh the spindle box has instead uh
like sort of the opposite aim in this
case the child has in front of them a
box with already uh some uh the serious
of number uh numbers impressed on the
box and with 45 spindles all identical
 $45 = 9 + 8$ plus seven Etc and the the work
consist in taking the spindles out from
the from the box and putting them back
according to the in in a way that the
number of spind of of spindles in each

um in each compartment has to um to be the same that is represented uh by the symbol itself

um as we see here the series of number goes from 0o till 9 in this particular exercise compared to the other one we will have just one extra element which is the child has to understand that in a series of number zero represent no quantities so it has the same so it somehow reinforces the uh the knowledge that has previously acquired with the other uh with the other exercises and now adds adds up with an over element the third uh exercise in m matics is the numerical coin in this case that it works as sort of test for children of three four years of agage in which uh the children have not only to combine the symbols with the quantities but also to put the symbols in the right order of course there is no shame in going back and in doing the different exercise at different levels and this is this is very important

because as we uh see here this um I'm sorry that I Ed the Italian but after having had the very bad experiences with the the English translations I now uh I tend to use directly the Italian um uh this this page refers actually to the different uh calculations that can be uh done using this kind of uh this kind of material at different stage of development if a child of three or four um might do the basic arithmetics of $9 + 1$ $8 + 2$ Etc noticing how um $9 + one$ so

um combining the road of nine plus the road of one you have the same uh measure the same Dimension that the road of 10 so it's equal 10 a child a little bit uh older that had already some experience with other forms of calculation for for instance multiplication so my notice that uh combining the uh the roads you might uh have uh five roads all with the same the same length the length of the 10 roads and you will have left the five roads in this way making a different calculation so calculating the total um number the the total quantity total uh number of units in this series which is 10 multiplied five + five which at the end gives M the possibility at again a different stage to calculate so gives the basics for calculating the um complete sum the complete quantity of uh um the complete quantity within a series of given numbers okay and I swear I am about finish these are uh three of the three other um elements of the extensive curriculum in mathematics and science it's not a case that the motasi um method seems to have the best result in mathematics and science because the the curricul the materials for mathematics and science particularly for the um the first grades is the most extensive one um now I'm not describing in in detail all these material just the the pearls that are the one on the on the left of the screen are used for making a little

bit more complex calculation and also
for teaching to uh visually teaching the
uh the poils how to calculate cubes
volumes
um but some times ago I uh attended to
this beautiful uh talk of Anthony trory
that is also an implementation in school
for the study of Computer Sciences in in
primary education um through a concept
that is called the concreteness fading
so these are implementation for
understanding how to uh help uh poops
particularly young children to construct
abstract knowledge particularly for very
abstract uh for very abstract
competencies such as uh computer
science uh the computer The concreteness
Fading approach practically consist
always in three moments for which we
start from a concrete element in this
case is the dead man island is is a sort
of game that the the kids have to
perform an intermediate phase in which
you add uh to uh some some elements of
the concrete phase some elements that
will be in the more abstract phase for
phase for then finally end up with the
obstruction this kind of concreteness
fading approach is everywhere in a mono
classro and it has the advantage that is
not just an implement mentation because
the implementations are useable and
beautiful the problem is that once the
implementation is brought away we have
the classroom as it stay as as it was
before instead in this case this kind of
um this kind of uh this these elements
are present in the classroom at all time

and this also helps for um uh
um somehow solving some problems with
this approach to uh for helping children
in constructing abstract knowledge which
is as always in in science in literat
particular in in literat education some
people think that the best way to help
kids in constructing abstract know
knowledge is going from concrete to
abstract some people think exactly the
opposite to present first the abstract
the abstract um knowledge and then to
allow the kids to make connection
between this abstract knowledge to the
uh to the uh concrete elements in in um
in everyday
word um we can assume very easily using
again an activ inference point of view
that both approaches might be right very
much depends on the kids having this
physical elements all the time into the
environment allow different kids with
different generative models to construct
their own um their own knowledge in
their own way some kids might find easy
to go from concrete to abstract some
kids might find easy to go from abstract
to concrete some kids might find easy to
go from intermediate to abser concrete
but anyway they will always be
surrounded by the same objects for
extensive periods of time and this allow
also to uh refine and to maybe uh going
from one to the other Approach at
different stages of
development um now the mon of course is
not the only uh valuable method of
Education that exists in the world this

is one of the the ones that I I like the most it's called the learning by the sign it has a different um different voices in in educational literature uh this is just one of them and I particularly like because uh things to um um change or to improve educational settings not from the head but from the feet uh so not through um educational pedagogies but really and physically rebuilding the uh environments in which the children or the poop learn every day the problem with this and other approaches of this kind is that they are not particularly precise in my opinion in looking at the error learning Dynamics and if we don't pay attention to uh um uh what a a poil perceives as an error what is gives the the error signal to the generative models of course the learning is way more difficult and this way more uh more complicated and at the end of the day when we want to design educational uh um educational environment we want any way to transmit certain kind of uh cultural and uh cultural competencies and the cultural knowledge finally um um we also haven't dedicated much at the end much thoughts about lms the the point with LMS is that it's clear that they are here to stay and is somehow recognized that they will have a huge impact on education the problem is how do we Implement them if we Implement them in a sort of traditional educational settings they will do for

the point what teachers do which means they will provide knowledge but we know that the knowledge that is provided by ILMs is not always accurate is sometimes messed up it is mixed up sometimes it is clearly invented therefore it's if we want to really use them within educational settings and we should because they are powerful entities um we should do that with what

Kasal have found a clear educational approach pedagogical approach uh with a strong focus of critical thinking and strategies something as the Montessori method finally the idea that I have that we have is more or less to combine the elements some elements that are taken from the Montessori from the Montessori method particularly for what concerns the uh the application in educational settings because this method already has in focus uh not only the child but also the teacher and also the curricula which is always um somehow regenerated by the interaction between the child and the teacher is I think maybe I should have mentioned earlier in the Montessori in the Montessori method the um interaction between teacher and pupils is seen as a form of collaboration not as a form of giving some kind of information and also that as in the focus is in the center of the focus at least a part of the extensive human knowledge that we have created in millennia compared with um uh combined with an approach of

learning to learning and uh to cognition
which is given by thetitive inference
which is a a predictive approach to uh
um the relationship between a cognitive
system and its environment which in my
opinion at least might be really the
game Cher in education when we will stop
to think about school as a as a place in
which the children are and the poop are
filled up with information but a place
in which they will um they
will try out what they already know and
they will refine their knowledge uh this
will be uh really a I think a game
changer in in educational
settings and that's all let me just
thank uh Ain carlut Ben White Axel
constant Andy Clark of course are part
of my team at the University of s Sussex
the chat lab the children and Technology
lab SE at the University of s Sussex
with the wonderful Nicola U and the
Escape material Minds year C grant that
pays my salary and allows me to uh uh
work on different aspects of materiality
for which I can work uh from educational
aesthtics to um to Tool making
activities in early home this is uh this
is really fun thank you very
much thank you Laura awesome
talk cool all right I can stop sharing
the screen on here all
right thank you um to kick things off
AAL if you want to give any first
Reflections and questions and then
meanwhile if people have any questions
in live chat I'll look for
that yes thank you Danielle so thank you

talk I it's clear it works good thing to
have in your

talk uh something I wanted to say in
addition resonance with what you said is
that you were my opinion very correct
point out that uh like an easy way to do
things would be to say that activ fance
tells us that we need to learn in know
certain active way that is not what is
happening AC inference is AAL model of
cognition if it's true cognition is
predictive that's it and whatever we can
do will not change

that um so AC inference does not tell
you that mon is better or that theal
system is unnatural that is not
something that happens but um what you
did and what I'd like to dig uh well in
the general same intuition but you know
different framework way of thinking is
that while it does not tell you that one
way to learn is better it does tell you
that well learning is contextual like
you if you learn to predict things you
learn to predict specific things from
specific
stimuli

so the way that you will uh learn things
will educate what type of stimuli you
give attention to and what type of
meaning uh you give to and uh the like
the last 20 400 years uh they were
characterized by a very very strong
cultural selection pressure for
institutions that could control people
doing things at very large scale so
mostly nation states mostly with war and
so they have a very strong interest to

for people to understand things through
the basically what the high things of it
and it happens that education system
Trends you to worry not about what it is
that you're doing but what the higher up
things you should think about what's
doing and if we
are if we are lucky uh possibly we may
uh have what is already occurring I
guess um well that is more and more
centered around like problem solving we
we we are a city we need to do something
with see what should we do and this is
an open question and your hup is not
like uh simultaneously a sociologist uh
Way Treatment manager
physicist so you'd have to have some
kind of active Demar of trying to find
out what it is that we should do and the
mon does that it does uh teach people to
like okay you have a problem here is the
environment new problem is set go ahead
do what you want and this trains people
that are I guess I hope more efficient
in problem solving that people who are
basically trying to learn the formula
and repeat it and then perform the
repeating formula so I think there is a
pretty deep thing to be the that and I
think you've identified it and I think
that there is like it's interesting to
think of it in terms of active inference
and there is more way to go uh in
understanding what precisely in terms of
active Insurance um sorry in by there is
more work to do in understanding in
terms of inference how precisely montori
or classical school system shapes uh the

way that people think and
organize can I can I answer yeah uh I
completely agree with you
obviously but um and maybe I wasn't PR
enough so the point is that as you said
activ if this is the way in which if
this is the way in which our cognitive
system works it will work in the Montes
meod in The Valor Shou in in whatever
system it will
be the problem is how much in the moment
in which our intent is
educative how much we are precise in
thinking the environment and the precise
context of
learning for being sure that the poops
will acquire the specific information
that we want them to acquire which
doesn't mean uh that we want all the
kids to think the same but we want for
instance to transmit a certain level of
basic competencies basic mathematics
basic literacy right from which then
they can build up by
themselves for doing that I called them
here because I didn't have a a better
term error learning Dynamics because the
point is uh as you said when when a
child goes to a traditional school or
whatever I I am personally against
grades and notes and all these kind of
stuff because they are um there's a
bunch of literature that say how bad
they are for motivation of the kids but
for me they have also another uh side
effect first they are delayed in time
which means if you think in in real
activ inference there are posteriors

that never refer to what they should
refer to if I have a bad grade of a test
that for a test that I've done last
week okay I received my error signal
doesn't come from the test doesn't come
from the mistakes that I've made in the
test it comes from the red notes that I
received from the teacher what I will
try to fix in my environment and how I
will try to update my generative model
is going to be oh I want to avoid that
this is my prediction error my my
attempt on the long run to solve
uncertainty right but this doesn't fit
with the idea that we want the child or
the poils or whatever or even the the
young adults to
learn the content that was in the test
in the first place this is the the this
is the in my opinion the main big
problem the second problem which somehow
the mono solved but the inference is
better is how we think how we have to
design
learning because if you
think learning as something that
is uh that Fs in the brain of the
child it's thinking in active inference
terms is in my opinion even more precise
that thinking in terms of construction
of knowledge right because the knowledge
is already somehow constructed you
arrive and you try it out the knowledge
that you have and then you refine and
refine and refine is this approach that
would I think could help in design
learning environments Learning Materials
how we design the books how we design

the uh interface how we design the the
team group they have to be designed in
this way a child that wres think about
the the basic I I try to think in very
very practical terms uh a child that
arrives to school is six and has to
learn to read and write and the first
thing this is a the mon meod on thisit
like it's it's it's okay because
theoretically you teach literacy at very
early age so you make the children play
around with letters at the age of three
or four they literally play around with
big letters but theoretically when you
go to school and you want to teach
children to uh read and write what do
you do the children arrive the first day
uh in class and they start with the
a tell me how can a cognitive system
predict something as such when a child
of
six knows how many words are connected
by the the the thing that you want to
learn think about how this is complete
nonsense in my opinion is complete
nonsense instead if you think the other
way around what does the child know
already the child knows Stories the
child know characters the child knows
structures the child knows context and
you think to use these as form of
predictions that the child will make and
every time you add one single element of
uncertainty manageable uncertainty
then it's going to be way
more way more
natural I'm sorry to say but that's it
that's is uh that's is how it

is okay I'll read a question next from
the live chat glea maximalist wrote what
is the monor perspective on peer
learning and social Clicks in the
classroom how does this relate to active
inference modeling of learning
um

okay

um the montori approach to
learning

is it's it's a little bit uh complex so
um in the Mont approach Learning and
Development they go together and you
have the phases of development which in
my opinion fits way better with the with
the activ inference that the the the the
of development of P or

bosi they are kind of rough periods of
development in which you construct
abstract

competencies uh but if you think in
specific terms monori think that the
first competence that the cognitive
system uh uh construct is the um
capacity of absorbing from the
word okay the this is the first thing
that a child does monor describes uh the
the uh the infant the newborn as
searching into the environment the
information that he can get it has very
much to do with our approach in activ
inference terms uh of of the model of
learning um albite of course is not so
precise first for uh

um historical

reasons uh she didn't have the right
terminology she didn't have the the
right knowledge uh she didn't have the

right uh neuroscientific knowledge and many of the things that montesori said or claimed they H had have been proved uh uh only in the past 50 years or so there is a beautiful book on montor Neuroscience written by Leonardo fasi the the guy of the mirror one of the guys of the miror neurons

um and for instance about the role of the hand in in cognition and imitation the role of the activities the role of perception they are already all of them inside the montor meod and they very much are in line with our approach our activ inference approach to learning and cognition

um I'd say that they are compatible okay that would be that I should have started with that they are compatible okay

um I hope that answers if it doesn't keep

going cool yeah well you made a very provocative comparison

with how the llms are trained with a corpus and a little bit of fine-tuning and then basically the content delivery instructional approach where there's a corpus there's the textbooks and the material and then with some fine-tuning we get to what a corpus regurgitating machine and along with that there's also these kind of implicit cultural learnings

not just related to the embodiment and the layout in the classroom but things like I should be predicting what

everyone else predicts or the next piece of information on my foraging Journey will be provided to me um or there is an expertise hierarchy that's inviable and all these different kinds of higher order properties that that become learnt through the education system and I think aval made a great point which is that active inference is more of the tool that helps us analyze these different cognitive settings and then talk about how they align or not or or how they fulfill our preferences or not um it's just uh it's very interesting to use this new

lens to bring together many of the the critiques that people have raised about education and I guess how do we move from taking the lens and applying it in a critical fashion to proactively designing new curricula and new experiences for for younger and for older

individuals I remember uh at some point I don't I don't remember whom but it was a Critic of

Education to said there is just one way to uh to reform education we have just to like to cancel schools that is the best case scenario I'm not so extreme but uh the idea to

change the idea to bring something like the T inference within educational settings means first and foremost not thinking about about education in terms of curriculum not thinking about education in terms of uh what a student

should know at the end of the year or
what a teacher should teach in one
year but is thinking about a
environment in which you Foster
options thinking about in in activ
inference terms means uh in a inference
a cognitive system what does
particularly a newb cognitive inst
in uh active inference uh um agent uh
try looks around simply looks around for
gathering information because they might
be useful for
making uh prediction error accurate
prediction errors in the
future which how do you translate this
into
education means offering
opportunities means offering as much as
it possible for making better prediction
errors right means
shifting completely the
mindset from what a person a child a
poop should
know to
what we can
offer what they can
know it's clear that's is for me it's
the mindset that I'd like to
shift yeah that's very important like
thinking about the adjacent possible and
the perception and the affordances of
each individual and that's also how
education and accessibility can come
together rather than trying to take like
a top- down organizational approach or
still again a Content delivery approach
just slower or diluted rather than
actually like welcoming people's

differences um okay Gia maximalist
writes what about the social structure
of the classroom first question and then
okay go for that and then I'll ask the
second question no no no h okay social
structure and and do you think that
grades are beneficial for older children
it reinforces that letter grades are the
delayed readout of performance like many
aspects of adult
life maybe we should take off the grades
also from adult's life that would be
make way easier social structure in in
the class in the montori classro the
social structure are very much again
natural so you have kids that forms
groups they can always
um work with whosoever they want so they
form sort of
natural friendships so you work more
with the people with whom you work
better so to say
um and the social structure of course I
couldn't go too much into detail um
about different social learning but if
you think how the uh the uh information
is provided Ed in a in a in a Mont
classroom the teacher never teaches so
you will never find a teacher that tells
you some stuff
and uh you have to repeat this stuff
most of the time particularly for basic
competencies the teacher
shows how to use certain materials or
helps you in uh finding information
online things
such this kind approaches and montori
was incredibly precise and really

sometimes when I read her really like
I'm astonished it was incredibly precise
saying how the teacher should talk with
with young children using gestures using
certain gestures in a way that they
could better vate the the message if I
think in terms of social learning and I
think about uh five or four years old a
six years old and a teacher which is a
adult competent individual and I think
in terms of social learning what does it
do stimulates over
imitation which is what is considered
the typical the most common of our way
to acquire information
socially which is also in
traditional schooling setting the one
that we never use and have you ever
tried to copy during a test it's the
first the first thing that you're are
supposed not to do but it's the most
natural way in which we transmit
cultural
information uh for saying about the
social structure this kind of different
mechanisms that again according to the
inference I see I see lots of advantages
to use active inferences for learning
because it allows to think the diff
kind of modalities in which we learn we
transmit and we learn
information uh cultur information among
each other uh as a society not as
separated in essence but more like a
sort of spectrum okay I try alone if I
don't
succeed then I will look I will shift my
attention to something else and so on

and so forth in a monor classroom this
is allow

children then can try out by themselves
if it doesn't work they can work
collaboratively uh collaboration if I
want to talk about not only social
structure but also the social learning
mechanism that are involved they work
collaboratively they imitate each other
all the time uh the imitation is not
simply um used like in in in in

Traditional High School you imitate the
cool kid in the corridor for how it's
dressed up you do the same in a Mont
classroom just you can work with them
and usually is not the child that is
dressed the better but the child with
whom you work better that is your cool
child in a classroom this changes a lot
the the

Dynamics uh grades grades grades grades
there is right now there is a big
movement against against grades I think
they have been published at least four
books in the past five six years against
grades the use of grades in in uh in
education

um

at for older kids the problem with
grades is that there is a gap between
whatever system we have for children in
compulsory school and when when we go to
university that is the main the main Gap
till the university works with exam and
grades and and so on and so forth we
will have to keep the grades into the
high schools because University allows
the access based on some

numbers that are given to you the later
we put the great the better
and the reason is very simple very young
children they
don't um they haven't yet developed the
ability to understand to make a
difference between what they do and who
they
are I have a bad note I am a bad person
okay
I it's very hard for them to make all
these levels these are all kind of
refinement of the generative model that
comes when you have multiple experiences
for which when you are 17 18 19 you
discover you start to discover that you
might be very bad at doing something but
you might be very good at doing
something
else for younger kids grades means a
sort of
identity that is the reason why I would
push them as back as it possible
honestly and there is also an ex
extensive literature in education
looking at how uh teachers should give
grades because they uh are really really
really bad for motivations really really
bad I can give you a personal experience
I was very good in chemistry till at
some point I got a bad grade because my
teacher discovered that I I wasn't uh
doing just my test but I was doing also
the test for two of my classmates and
instead of saying okay you are very good
you have been so good and so fast that
you could finish your own test and
making the test for other two

classmates she gave me a b grade and
never been in the class of chemistry
ever

again uh and I think as me since I'm not
special as me many other kids many other
uh students would feel exactly the same
yeah another kind of piece with the
language models and the synthetic
intelligence in the classroom yes
they're disembodied and there is that
angle that we explored also though it's
kind of like there's a new smart kid on
the Block there's a new affordance which
is like what year did this happen in and
that went from being something maybe
that could be found in a book more
actions that have to be taken to walk to
the library open the book Etc to maybe
now search for it online to maybe now
it's just going to be like clap your
hands and say it again or maybe it's
listening the whole time and so how will
the

meta on developing personal
agency and a feeling of being in the
driver seat in the education be
cultivated so people feel like they have
tools to access and they have
affordances at their
disposal rather than feeling I don't
know how it could or would feel but to
be under the burden of much more
knowledgeable much
faster agents that are directing the
students and teachers I think might also
throw up their hands because they're not
necessarily going to have recourse for
technical questions either and there

will be this affordance at hand that
will just bring out an essay or
increasingly the audio and the visual so
how do we have a kind of fusion
environments or different zones that
help us wisely use that
technology that's is a very
so I am a Sci-Fi fan Okay so I am not
scared about anything of this kind bring
in I can deal with that um honestly
after having been use having been using
for uh at this point one year
consistently large language models while
the at the beginning I thought okay this
is really a game Cher right now I think
um not so much at the end
um the point is our approach the point
is
how we teach people how to use them
and as for every other piece of
technology we all will use a different
in different way in different
circumstances if I have to uh to
translate I live in Germany if I have to
translate a piece that I'm writing for
sending to I don't know an mail to the
mother of uh my daughter of um my
daughter friend uh I can use J GPT and I
will send the translation as it is okay
I won't pay too much attention when I
use jpt for working instead not only I
check thousands of times the information
that I have received
because despite it might have faster
access to the knowledge but doesn't
have the right knowledge
somehow and this is something that
you

feel when it's right and when it's not
when you write a piss and you make J P I
make J P checking my grammar and my
mistakes because I'm not mother pong
when gives me back I always pay
attention to put just no more than 10
lines because otherwise restructures my
sentence and usually what comes out is
nonsense is changes completely the
meaning and stuff as such so uh yes they
are useful we have to learn how to use
them as we have learned to use uh to use
I remember in
2011 yes was 2011 came out a paper I
think it's called the Google effect and
they looked at how much we have uh uh in
in in the opinion of the of
the uh research Searchers we uh we have
been losing memory because we started to
use Google all the time instead of
memorizing things right uh I see things
in a different way uh we haven't lost
memory we have learned how to categorize
things that we want to remember in a
different way I don't have to store the
information itself I have to
remember how to retrieve that
information and the same will be for jpt
and for the large language
models also sometimes ago I was I was
using jpt and at some point and this is
the way in which the moment in which it
was clear to me um I asked for some uh
references about some tool making stuff
of for earo and it provided me with a
book of uh Kim sery uh um of a book of
Kim sney and uh Beyond I know Kim
perfectly I know his work very very very

nicely and I looked at this book I said
but this book doesn't exist I'm 100%
sure that this book I I think I I looked
at the title I think this is the book
that he should write but this book
doesn't exist because at the end of the
day I know
better nice yeah many funny things like
that and and knowing where to find the
information whether the internal
foraging or external Niche modification
it's kind of always been the game it's
just a little bit modified now um now
I'm sorry interrupt you I'm now working
on uh um txm on extended mind and
learning and the active inference here
has a very nice way to uh to change the
game of uh extra ST ding of information
and so on she's very very interesting
yes I have a friend who says he has the
largest collection of seashells in the
world he leaves them on all the
beaches um well where do you go from
here or what would you like to
see happen or what would you suggest to
people who are excited about pursuing
education and active
inference uh I don't know because I I as
I said I I tried to to to create with
jpt I would I would show to you all the
attempt uh to create a um image um of a
classroom in which was applied the
theory of cognition of activ infant with
the Mont method and I
had lots of weird results because it's
something that apparently nobody has
tried
yet but what I would suggest is trying

to make uh practical to think in terms
of design to think in terms of
affordances to think in terms of
everyday interactions to think in terms
of buildings to think in terms of
artifacts this I think is going to be
the The Next Step
awesome yeah just to read a few comments
Joanne wrote it brings the environment
into the picture we need to examine that
more closely collaboration not
competition and then Jr wrote It's like
lifting memory from remembering the
object to remembering the process in a
more complex context as the world
evolves yes yes the the the good aspect
I think the point of putting together
something like the monor meod and active
inference
is understanding that learning is not
about acquiring information but about
understanding the information how to use
the information that is the the the game
changer or as my daughter says all the
time Mama today I invented
mathematics amazing well Laura thank you
very much for the work it was definitely
a topic
that people I feel like wanted and want
and need to explore how to be active in
education and not get swamped in the
tools and in all of these efforts and to
kind of keep the hope alive for what
education can
be yes yes if you should put a little
bit more uh ecology and sustainability
inside and then we have the complete
picture awesome well till next time

thank you very much next time thank you
very much bye bye